

MATH 2551-K MIDTERM 3

VERSION A

FALL 2023

COVERS SECTIONS 15.5-15.8, 16.1-16.8

Full name: _____

GT ID: _____

Honor code statement: I will abide strictly by the Georgia Tech honor code at all times. I will not use a calculator. I will not reference any website, application, or other CAS-enabled service. I will not consult with my notes or anyone during this exam. I will not provide aid to anyone else during this exam.

() I attest to my integrity.

Read all instructions carefully before beginning.

- Print your name and GT ID neatly above.
- You have 75 minutes to take the exam.
- You may not use aids of any kind.
- Please show your work.
- Good luck! Write yourself a message of encouragement on the front page!

For problems 1-3 choose whether each statement is true or false. If the statement is *always* true, pick true. If the statement is *ever* false, pick false. Be sure to neatly fill in the bubble corresponding to your answer choice.

1. Suppose $\mathbf{F}(x, y)$ is a curl-free vector field ($\nabla \times \mathbf{F} = 0$) defined in all of $\mathbb{R}^2$ except $(0, 0)$. Then $\mathbf{F}$ is conservative.

☐ **TRUE**
☐ **FALSE**

2. There exists a non-constant vector field $\mathbf{F}(x, y)$ with both $\nabla \cdot \mathbf{F} = 0$ and $\nabla \times \mathbf{F} = 0$.

☐ **TRUE**
☐ **FALSE**

3. If $\mathbf{F}$ is any vector field and C is a circle, then the circulation of $\mathbf{F}$ around C traversed clockwise is the negative of the circulation of $\mathbf{F}$ around C traversed counterclockwise.

☐ **TRUE**
☐ **FALSE**

4. Consider the double integral

$$\iint_R \frac{2x - y}{3y - x} dA,$$

where R is the region bounded by the lines $2x - y = 0$, $2x - y = 4$, $3y - x = 0$, $3y - x = 2$. Which of the following coordinate transformations would be appropriate to use for change of variables on this integral?

- ☐ $u = \frac{y}{x}$
 $v = \frac{x}{3y}$
☐ $x = u$
 $y = v$
☐ $u = 2x - y$
 $v = 3y - x$
☐ $x = 2u - v$
 $y = 3v - y$

5. Compute the flux of the vector field $\mathbf{F}(x, y, z) = x\mathbf{i} + y\mathbf{j}$ through the open-ended circular cylinder of radius 4 and height 7 with its base on the xy -plane and centered about the z -axis, oriented away from the z -axis.

6. In this problem, you will work with the path C which is the ellipse $\frac{x^2}{2^2} + \frac{y^2}{5^2} = 1$ oriented counterclockwise.

(a) Give a parameterization $\mathbf{r}(t)$ of the ellipse.

(b) Use any method to find the flux of the vector field $\mathbf{F} = \langle e^y, 2\cos(x) \rangle$ out of the interior of C .

(c) Use any method to find the circulation of the vector field $\mathbf{F} = \langle 2y, x \rangle$ around C .

7. Let $\mathbf{F}$ be the vector field

$$\mathbf{F}(x, y, z) = \langle 3x + e^{z^2 y}, y + z, \arctan(x) - z \rangle.$$

(a) Let S be the surface defined by

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{6}\right)^2 + \left(\frac{z}{2}\right)^2 = 1,$$

oriented with outward pointing normal vectors. Compute the flux of $\mathbf{F}$ across S .

Hint: The volume of the ellipsoid $(\frac{x}{a})^2 + (\frac{y}{b})^2 + (\frac{z}{c})^2 \leq 1$ is $\frac{4}{3}\pi abc$.

(b) Is there a vector field $\mathbf{F}$ such that $\mathbf{F} = \nabla \times \mathbf{F}$? Explain your answer.

8. Consider the volume D in the second octant of $\mathbb{R}^3$ ($x \leq 0, y \geq 0, z \geq 0$) which is bounded above by the sphere $x^2 + y^2 + z^2 = 2$ and below by the paraboloid $z = x^2 + y^2$.

(a) Write an integral for the volume of D using Cartesian coordinates in the order $dz \, dy \, dx$. Show your work. **Do not evaluate your integral.**

(b) Write an integral for the volume of D using cylindrical coordinates. Show your work. **Do not evaluate your integral.**

Consider the volume D in the second octant ($x \leq 0, y \geq 0, z \geq 0$) of $\mathbb{R}^3$ which is bounded above by the sphere $x^2 + y^2 + z^2 = 2$ and below by the paraboloid $z = x^2 + y^2$.

(c) Sketch the shadow of D in the yz -plane.

(d) Are spherical coordinates a good choice to use to compute this volume? Justify your answer.

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FORMULA SHEET

- Trig identities: $\sin^2(x) = \frac{1}{2}(1 - \cos(2x))$, $\cos^2(x) = \frac{1}{2}(1 + \cos(2x))$
- Volume(D) = $\iiint_D dV$, $f_{avg} = \frac{\iiint_D f(x, y, z) dV}{\text{volume of } D}$, Mass: $M = \iiint_D \delta dV$
- Cylindrical coordinates: $x = r \cos(\theta)$, $y = r \sin(\theta)$, $z = z$, $dV = r dz dr d\theta$
- Spherical coordinates: $x = \rho \sin(\phi) \cos(\theta)$, $y = \rho \sin(\phi) \sin(\theta)$, $z = \rho \cos(\phi)$, $dV = \rho^2 \sin(\phi) d\rho d\phi d\theta$
- First moments (3D solid): $M_{yz} = \iiint_D x\delta dV$, $M_{xz} = \iiint_D y\delta dV$, $M_{xy} = \iiint_D z\delta dV$
- Center of mass (3D solid): $(\bar{x}, \bar{y}, \bar{z}) = \left(\frac{M_{yz}}{M}, \frac{M_{xz}}{M}, \frac{M_{xy}}{M}\right)$
- Substitution for double integrals: If R is the image of G under a coordinate transformation $\mathbf{T}(u, v) = \langle x(u, v), y(u, v) \rangle$ then

$$\iint_R f(x, y) dx dy = \iint_G f(\mathbf{T}(u, v)) |\det D\mathbf{T}(u, v)| du dv.$$

- Scalar line integral: $\int_C f(x, y, z) ds = \int_a^b f(\mathbf{r}(t)) |\mathbf{r}'(t)| dt$
- Tangential vector line integral: $\int_C \mathbf{F} \cdot \mathbf{T} ds = \int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$
- Normal vector line integral: $\int_C \mathbf{F}(x, y) \cdot \mathbf{n} ds = \int_C P dy - Q dx = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \langle y'(t), -x'(t) \rangle dt$.
- Fundamental Theorem of Line Integrals: $\int_C \nabla f \cdot d\mathbf{r} = f(B) - f(A)$ if C is a path from A to B
- Curl (Mixed Partial) Test: $\mathbf{F} = \nabla f$ if $\text{curl } \mathbf{F} = 0 \Leftrightarrow P_z = R_x, Q_z = R_y$, and $Q_x = P_y$.
- $\nabla = \left\langle \frac{\partial}{\partial x}, \frac{\partial}{\partial y}, \frac{\partial}{\partial z} \right\rangle$ $\text{div } \mathbf{F} = \nabla \cdot \mathbf{F}$ $\text{curl } \mathbf{F} = \nabla \times \mathbf{F}$
- Green's Theorem: If C is a simple closed curve with positive orientation and R is the simply-connected region it encloses, then

$$\int_C \mathbf{F} \cdot \mathbf{T} ds = \iint_R (\nabla \times \mathbf{F}) \cdot \mathbf{k} dA \qquad \int_C \mathbf{F} \cdot \mathbf{n} ds = \iint_R (\nabla \cdot \mathbf{F}) dA.$$

- Surface Area = $\iint_S 1 d\sigma$
- Scalar surface integral: $\iint_S f(x, y, z) d\sigma = \iint_R f(\mathbf{r}(u, v)) |\mathbf{r}_u \times \mathbf{r}_v| dA$
- Flux surface integral: $\iint_S \mathbf{F} \cdot \mathbf{n} d\sigma = \iint_S \mathbf{F} \cdot d\boldsymbol{\sigma} = \iint_R \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA$
- Stokes' Theorem: If S is a piecewise smooth oriented surface bounded by a piecewise smooth curve C and $\mathbf{F}$ is a vector field whose components have continuous partial derivatives on an open region containing S , then

$$\int_C \mathbf{F} \cdot \mathbf{T} ds = \iint_S (\nabla \times \mathbf{F}) \cdot \mathbf{n} d\sigma.$$

- Divergence Theorem: If S is a piecewise smooth closed oriented surface enclosing a volume D and $\mathbf{F}$ is a vector field whose components have continuous partial derivatives on D , then

$$\iint_S \mathbf{F} \cdot \mathbf{n} \, d\sigma = \iiint_D \nabla \cdot \mathbf{F} \, dV.$$

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